Quantitative EEG and ERP report

CLINIC C LOGO

Patient:	Mock Patient 042
Date of birth:	1 Jan 2016
Age:	10 years
Sex:	female
Recorded:	15 Sep 2026
Case code:	MOCK-042
Earlier measurements:	23 Sep 2025 and 12 Mar 2026

This report presents the EEG, ERP and VCPT measurements of Mock Patient 042, aged 10, recorded on 15 Sep 2026.

Report	R-MOCK-042-0915 v1 · Draft, not issued	Recording system	Mitsar amplifier, WinEEG file (mockup label)
Template	CT-STD v4 · Clinic C · English	Result set	RS-MOCK-042-0915-1
Software	V4 mockup 0.1	Marking	Benjamini-Hochberg, q = 0.05 within each analysis block (stand-in)
Device identifier (UDI-DI)	UDI: placeholder, assigned at certification	Input checksum (sha256 of the de-identified signal)	`90a3d80f627a22c2564cc31c7a4196242572fcecff4a7d9729e3e283a83a3ef9` (synthetic input, not a recording file)
Pipeline	mockup pipeline 0.1.0	ICA seed	33226 (derived from the recording and stored)
Norms	Mock norms 2026.1 (synthetic); dataset version 2026.1	Processed	15 Sep 2026, 10:46, in 400 s
Text library	2026.2		
Device blocks	Device blocks 2026.2		

I. Summary

The largest marked deviations from the age norm in this measurement are: beta1 coherence, eyes closed: T3-T5 (z = 4.8), T5-P3 (z = 4.6), T5-O1 (z = 4.6), P3-O1 (z = 4.2), T3-C3 (z = 4.1), C3-T5 (z = 3.9), C3-P3 (z = 3.9); SMR asymmetry index, eyes open: C3-C4 (z = 4.3); beta1 coherence, eyes open: P3-O1 (z = 4.2), T3-C3 (z = 4.1), T5-O1 (z = 4.1), T3-T5 (z = 3.8), C3-T5 (z = 3.8), C3-P3 (z = 3.7), T5-P3 (z = 3.7); N1 amplitude, Go: O1 (z = -4.2), O2 (z = -4.2); SMR asymmetry index, eyes closed: C3-C4 (z = 4.1); beta2 coherence, eyes open: P4-O2 (z = 3.9); relative theta power, eyes open: P3 (z = -3.5), P4 (z = -3.4), O1 (z = -3.4), O2 (z = -3.3), Pz (z = -3.2).

This list ranks the marked values of these analyses by |z|: absolute and relative band power, maps per 1 Hz band, alpha peak frequency, theta/beta ratio, frontal alpha asymmetry index, frontal theta / parietal alpha ratio, alpha reactivity, asymmetry map, magnitude squared coherence per electrode pair, ERP peak amplitude and latency, and VCPT task results. It shows the 5 largest and every marked value of 3 z or more. Not ranked: the spectra per 0.5 Hz bin, shown as curves against the norm, global coherence in its four types, and lagged coherence per electrode pair. Source power per brain region is not part of this list.

Marked values in this measurement: band power 5 of 456, maps per 1 Hz band 0 of 1,102, alpha peak and ratios 0 of 9, asymmetry map 2 of 96, magnitude squared coherence 15 of 2,052, ERP peaks 2 of 21, VCPT task 0 of 4, spectrum per 0.5 Hz bin 0 of 2,242, global coherence 0 of 48 and lagged coherence per electrode pair 0 of 2,052.

Figure 1: Largest marked deviations, as in the summary

Analysis and condition	Electrode or pair, z
beta1 coherence, eyes closed	T3-T5 **4.8** T5-P3 **4.6** T5-O1 **4.6** P3-O1 **4.2** T3-C3 **4.1** C3-T5 **3.9** C3-P3 **3.9**
SMR asymmetry index, eyes open	C3-C4 **4.3**
beta1 coherence, eyes open	P3-O1 **4.2** T3-C3 **4.1** T5-O1 **4.1** T3-T5 **3.8** C3-T5 **3.8** C3-P3 **3.7** T5-P3 **3.7**
N1 amplitude, Go	O1 **-4.2** O2 **-4.2**
SMR asymmetry index, eyes closed	C3-C4 **4.1**
beta2 coherence, eyes open	P4-O2 **3.9**
relative theta power, eyes open	P3 **-3.5** P4 **-3.4** O1 **-3.4** O2 **-3.3** Pz **-3.2**



Reading key: z is the distance of a value from the normative mean in standard deviations of the normative sample. Cell and map colours show z on the fixed scale (blue below, red above the age norm). Bold with a ring: a value marked after correction for multiple comparisons. Dotted line in the maps: |z| = 2, which alone does not mark a value.

II. Signal quality and processing

Eyes open: 164 of 180 seconds were analysed after artefact removal.

Eyes closed: 171 of 180 seconds were analysed after artefact removal.

VCPT: 1168 of 1220 seconds were analysed after artefact removal.

Accepted VCPT trials: Go 95, NoGo 92, Ignore 93, Novelty 95 (of 100 each).

Artefact removal changed by the clinician: component 6 (muscle, left temporal) was kept although the automatic cleaning proposed removing it; 2 of 18 components were removed.

The analysed segments were confirmed by the clinician as proposed.

Alpha peak frequency, eyes closed (posterior mean): suggested 9.2 Hz, confirmed unchanged by the clinician; not marked (z = 0.0).

III. Anamnesis and clinical interview

Anamnesis	Synthetic example text. Seen for a quantitative EEG and ERP assessment at the family's request; third measurement at this clinic.
History	Synthetic example text. No relevant events reported by the parents.
Medication	Synthetic example text. None reported.
Current symptoms	Synthetic example text. Sleep and appetite as usual, as reported by the parents.

IV. Spectra and maps

Eyes open

Figure 2: Absolute power, z against the age norm, Eyes open

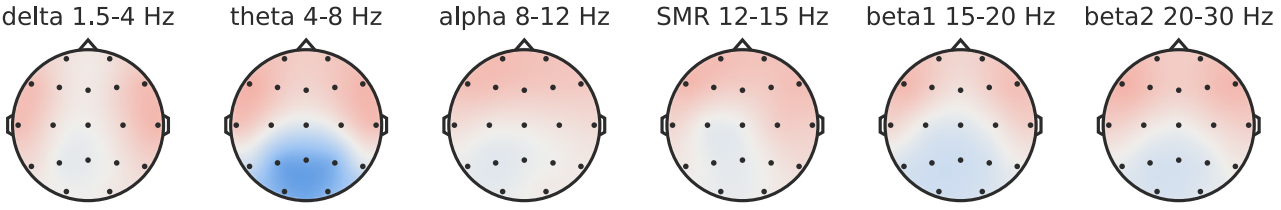
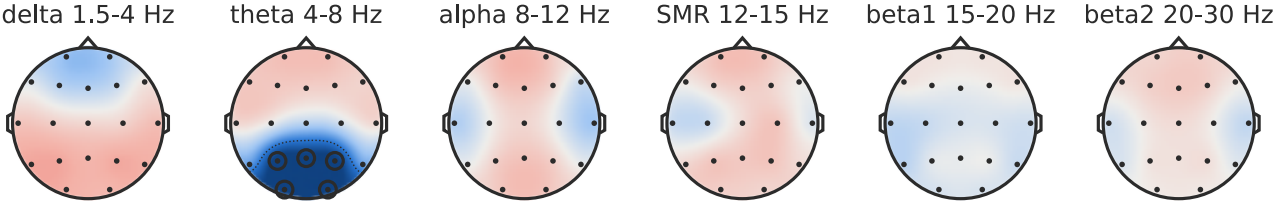


Figure 3: Relative power, z against the age norm, Eyes open



Relative theta power, eyes open, is below the age norm at P3 (z = -3.5), Pz (z = -3.2), P4 (z = -3.4), O1 (z = -3.4) and O2 (z = -3.3).

The alpha peak frequency, eyes open, is 9.1 Hz and is not marked (z = 0.0).

Eyes closed

Figure 4: Absolute power, z against the age norm, Eyes closed

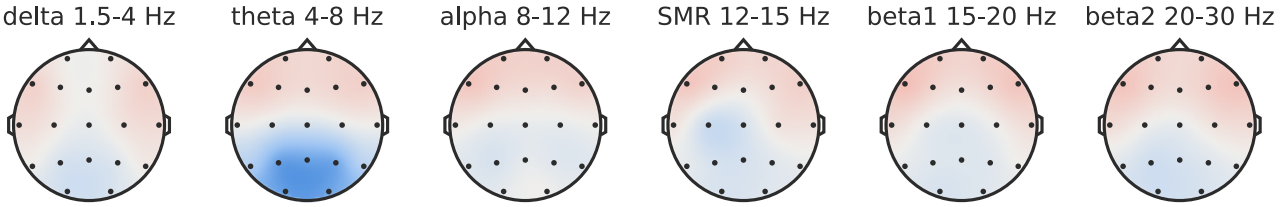
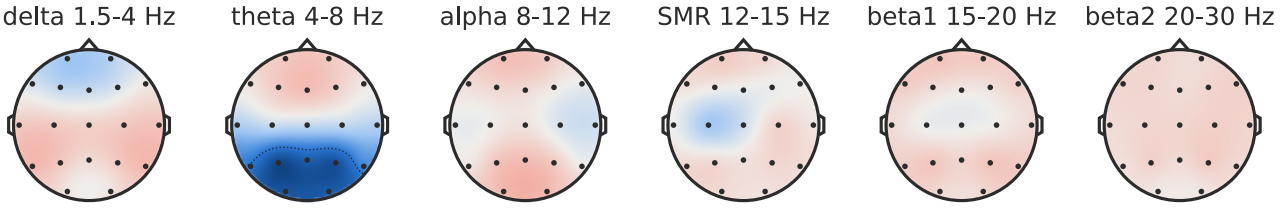
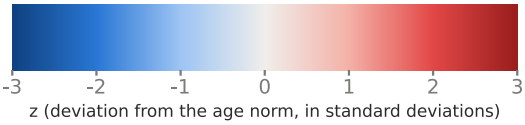


Figure 5: Relative power, z against the age norm, Eyes closed



No absolute or relative band power value, eyes closed, is marked.

The alpha peak frequency, eyes closed, is 9.2 Hz and is not marked (z = 0.0).



Ring: marked value (after correction for multiple comparisons).

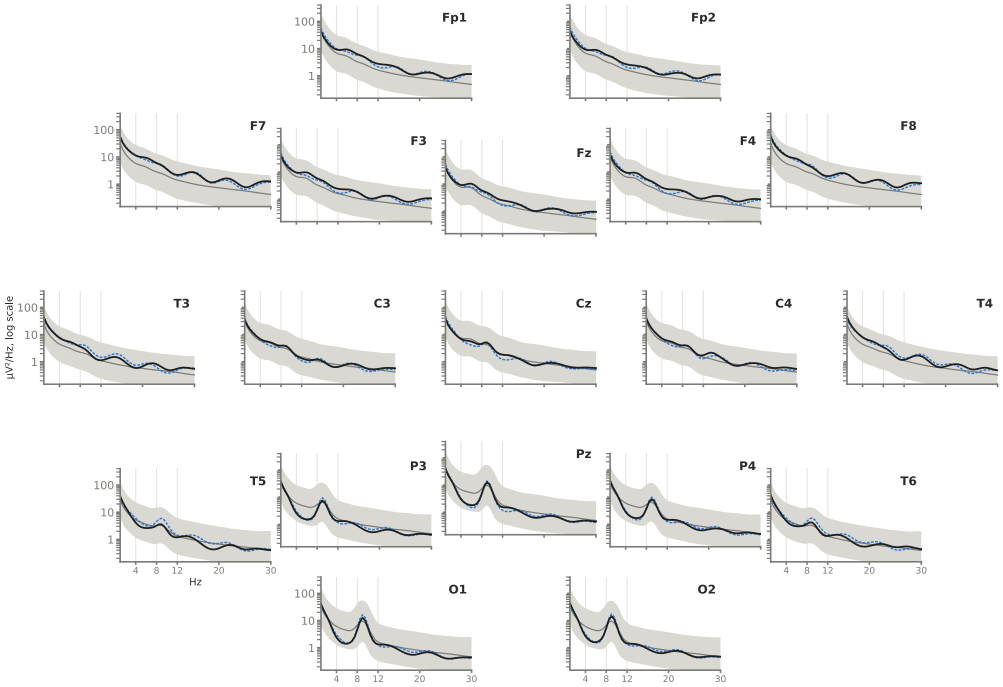
Clinician's comment

Clinician block, empty in this sample.

V. Absolute spectra per electrode

Eyes open

Figure 6: Absolute spectra per electrode, Eyes open



Eyes closed

Figure 7: Absolute spectra per electrode, Eyes closed

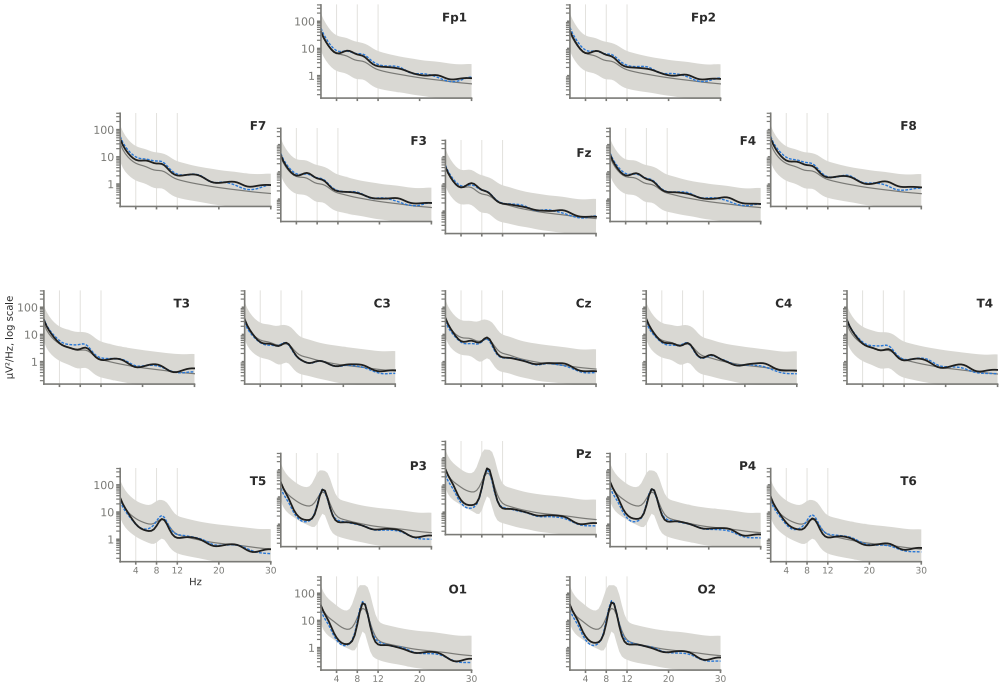


Figure 8: Relative spectra per electrode (share of total power per 0.5 Hz bin), Eyes open

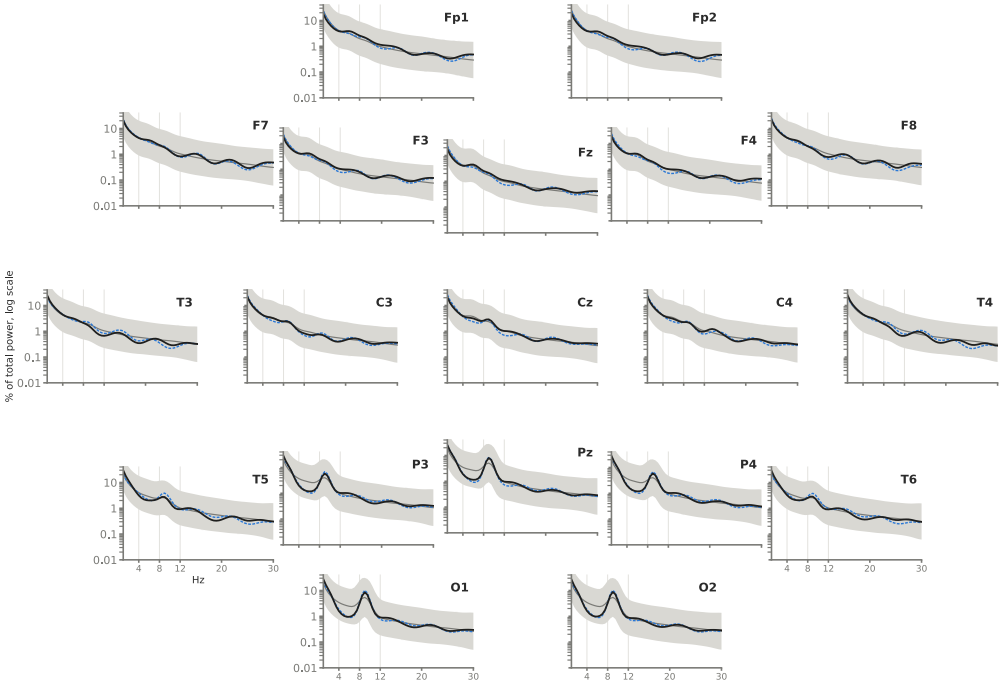
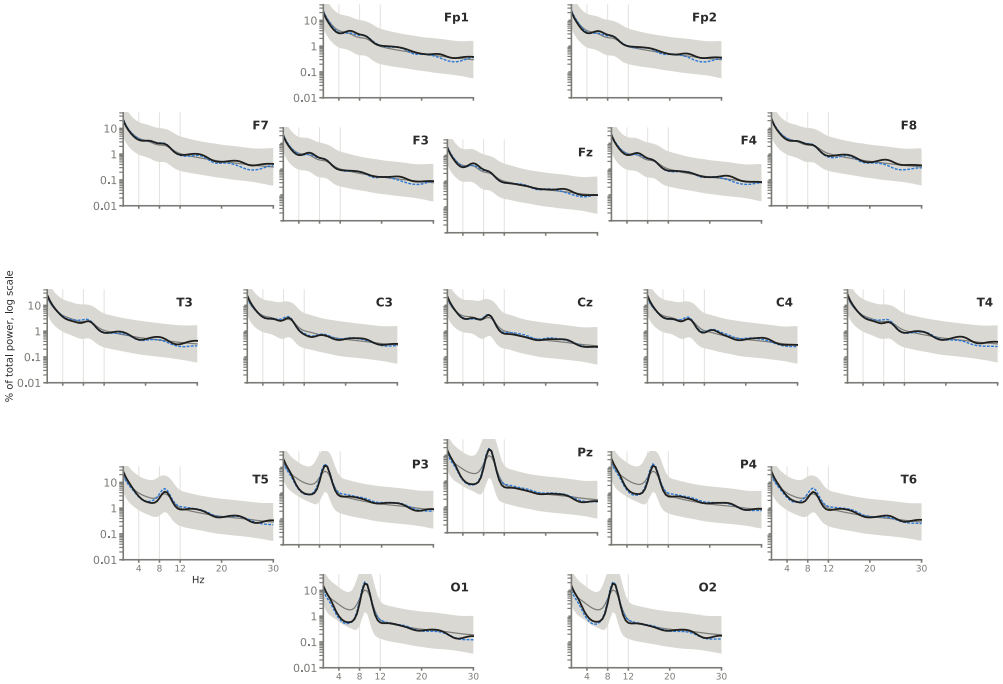


Figure 9: Relative spectra per electrode (share of total power per 0.5 Hz bin), Eyes closed



Black: this measurement. Grey line and band: normative mean and mean ± 2 SD. Blue dashed: earlier measurement. Norm band of the relative spectra and z of the relative maps per 1 Hz band: a stand-in derived from the absolute norm in this mockup.

VI. Frequency markers

Figure 10: Frequency markers

No.	Condition	Electrode	Frequency (Hz)	z	Question, answer	Choice
1	Eyes open	C4	13.5	0.7	-	suggested, confirmed
2	Eyes open	O2	9.0	0.5	-	suggested, confirmed
3	Eyes open	C4	21.5	0.4	-	suggested, confirmed
1	Eyes closed	F3	6.0	0.6	-	suggested, confirmed
2	Eyes closed	O2	9.0	0.5	-	suggested, confirmed
3	Eyes closed	C4	23.0	0.4	-	suggested, confirmed

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 11: z at the marker frequency, Eyes open

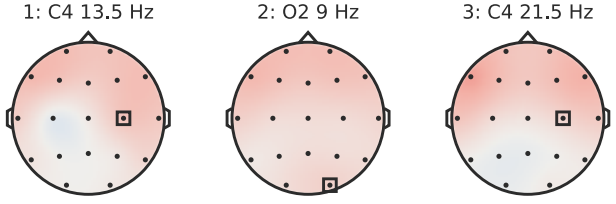
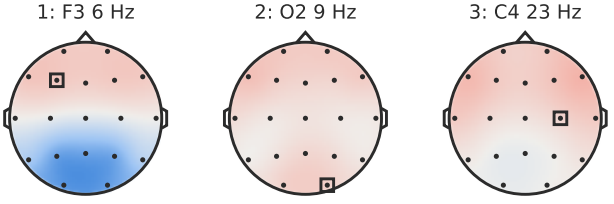


Figure 12: z at the marker frequency, Eyes closed



Square: the electrode of the marker. Ring: marked value (after correction for multiple comparisons).

Frequency markers, eyes open: suggested by the system and confirmed unchanged by the clinician; markers: 3.

Frequency markers, eyes closed: suggested by the system and confirmed unchanged by the clinician; markers: 3.

The system proposes up to 3 peaks of the spectrum per condition, those with the largest positive z (absolute power per 0.5 Hz bin); the clinician confirms, moves, removes or adds markers. Each marker shows its z and whether that value is marked.

VII. Maps per 1 Hz band (absolute power, z)

Figure 13: Maps per 1 Hz band (absolute power, z), Eyes open

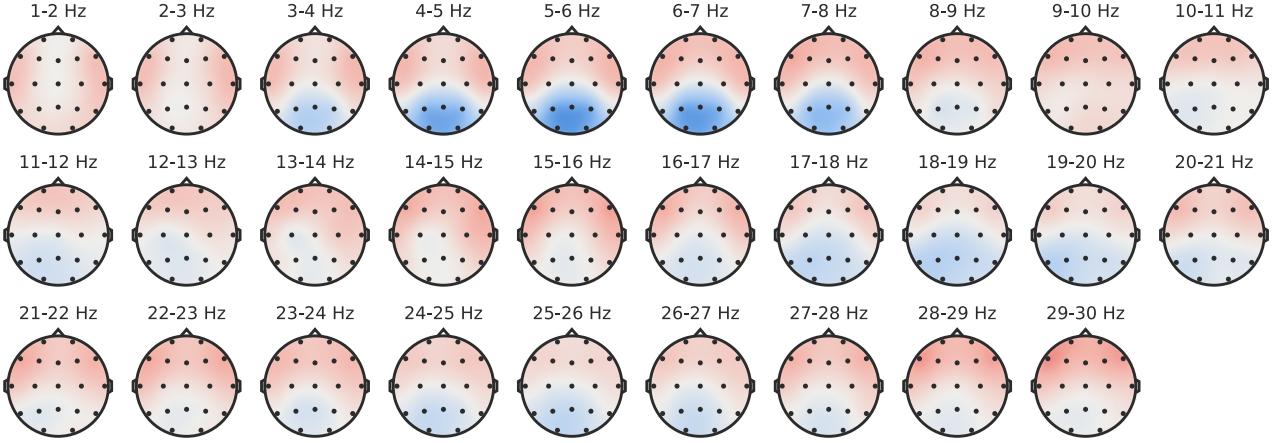
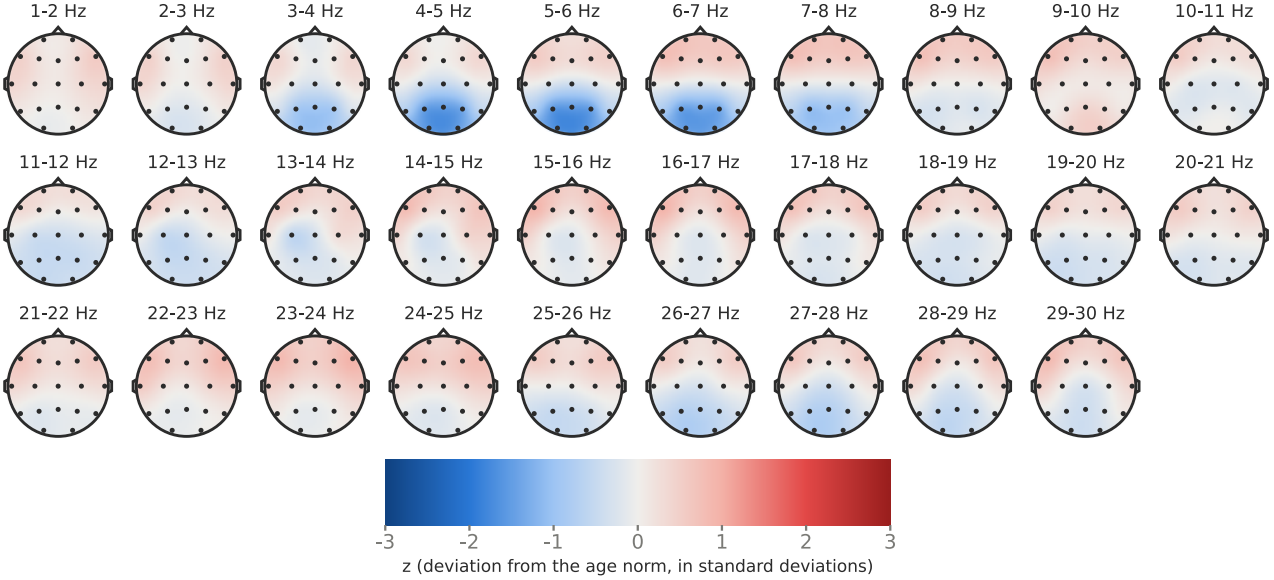


Figure 14: Maps per 1 Hz band (absolute power, z), Eyes closed



Ring: marked value (after correction for multiple comparisons).

VIII. Band ratios and indices

Measure		Value	Norm mean	z
Theta/beta ratio at Cz	Eyes open	2.54	2.76	-0.3
Frontal alpha asymmetry index (ln F4 minus ln F3)	Eyes open	-0.06	-0.01	-0.6
Frontal theta / parietal alpha ratio	Eyes open	1.78	1.11	0.7
Alpha peak frequency (Hz)	Eyes open	9.1	9.1	0.0
Theta/beta ratio at Cz	Eyes closed	2.66	2.82	-0.2
Frontal alpha asymmetry index (ln F4 minus ln F3)	Eyes closed	-0.07	-0.01	-0.6
Frontal theta / parietal alpha ratio	Eyes closed	0.72	0.46	0.7
Alpha peak frequency (Hz)	Eyes closed	9.2	9.2	0.0
Alpha reactivity	Eyes closed / eyes open	2.49	2.38	0.1

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Frontal theta / parietal alpha ratio: mean absolute theta at F3, Fz and F4 over mean absolute alpha at P3, Pz and P4. Alpha reactivity: mean absolute alpha at P3, Pz, P4, O1 and O2, eyes closed over eyes open. Both are compared with the norm on a log scale and printed as the ratio.

The theta/beta ratio at Cz, eyes open, is 2.54 (z = -0.3).

The frontal alpha asymmetry index (ln F4 minus ln F3), eyes open, is -0.06 (z = -0.6).

The frontal theta / parietal alpha ratio (mean of F3, Fz and F4 over mean of P3, Pz and P4), eyes open, is 1.78 (z = 0.7).

The theta/beta ratio at Cz, eyes closed, is 2.66 (z = -0.2).

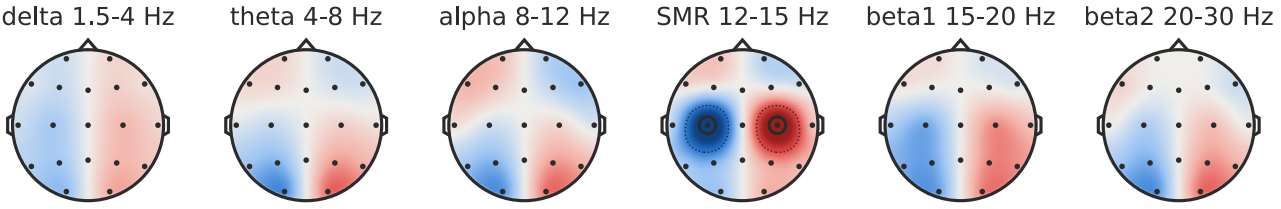
The frontal alpha asymmetry index (ln F4 minus ln F3), eyes closed, is -0.07 (z = -0.6).

The frontal theta / parietal alpha ratio (mean of F3, Fz and F4 over mean of P3, Pz and P4), eyes closed, is 0.72 (z = 0.7).

Alpha reactivity (absolute alpha power, mean of P3, Pz, P4, O1 and O2, eyes closed over eyes open) is 2.49 (z = 0.1).

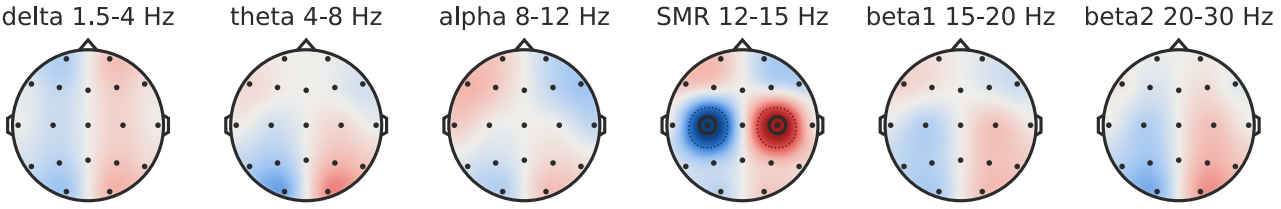
IX. Asymmetry

Eyes open



The SMR asymmetry index C3-C4, eyes open, is 0.24 (more power at C4), above the age norm (z = 4.3).

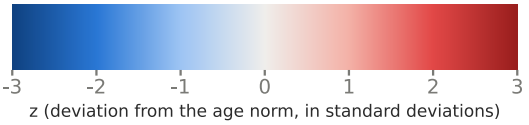
Eyes closed



The SMR asymmetry index C3-C4, eyes closed, is 0.25 (more power at C4), above the age norm (z = 4.1).

Figure 15: Marked electrode pairs

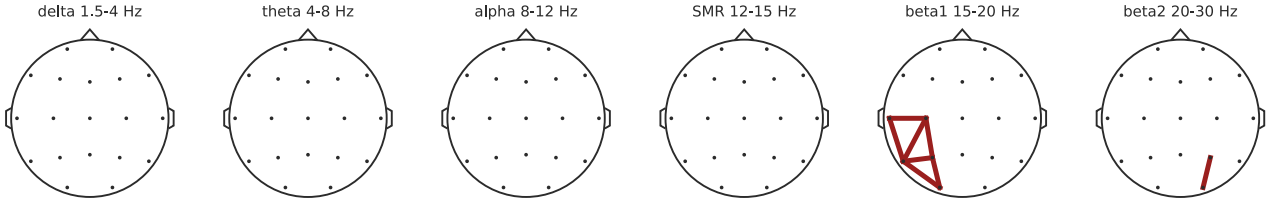
Condition	Band	Pair	Index	More power at	z
Eyes open	SMR	C3-C4	0.24	C4	○ 4.3
Eyes closed	SMR	C3-C4	0.25	C4	○ 4.1



Asymmetry index (right minus left) / (right plus left) per electrode pair: its z against the age norm at the right electrode of the pair and the same z with the opposite sign at the left electrode; midline 0. Ring: marked value (after correction for multiple comparisons).

X. Coherence

Eyes open, magnitude squared coherence

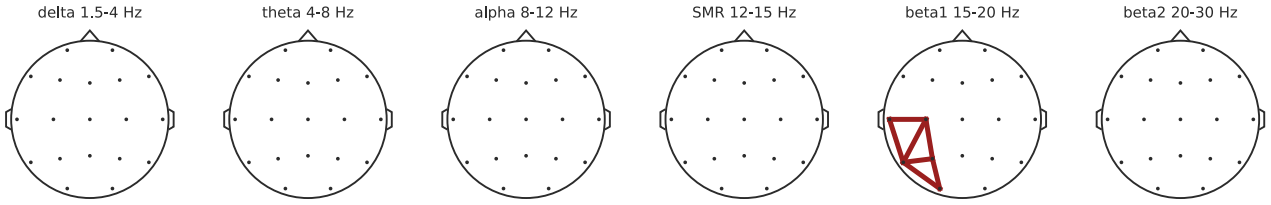


Beta1 magnitude squared coherence, eyes open, is above the age norm for 7 electrode pairs: P3-O1 (z = 4.2), T3-C3 (z = 4.1), T5-O1 (z = 4.1), T3-T5 (z = 3.8), C3-T5 (z = 3.8), C3-P3 (z = 3.7) and 1 more in the table.
Beta2 magnitude squared coherence, eyes open, is above the age norm for 1 electrode pair: P4-O2 (z = 3.9).

Figure 16: Marked electrode pairs, magnitude squared coherence, Eyes open

Band	beta1							beta2
Pair	P3-O1	T3-C3	T5-O1	T3-T5	C3-T5	C3-P3	T5-P3	P4-O2
Coherence	0.73	0.69	0.64	0.62	0.60	0.64	0.70	0.68
z	○ 4.2	○ 4.1	○ 4.1	○ 3.8	○ 3.8	○ 3.7	○ 3.7	○ 3.9

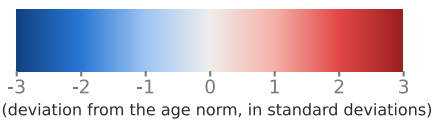
Eyes closed, magnitude squared coherence



Beta1 magnitude squared coherence, eyes closed, is above the age norm for 7 electrode pairs: T3-T5 (z = 4.8), T5-P3 (z = 4.6), T5-O1 (z = 4.6), P3-O1 (z = 4.2), T3-C3 (z = 4.1), C3-T5 (z = 3.9) and 1 more in the table.

Figure 17: Marked electrode pairs, magnitude squared coherence, Eyes closed

Band	beta1						
Pair	T3-T5	T5-P3	T5-O1	P3-O1	T3-C3	C3-T5	C3-P3
Coherence	0.68	0.76	0.67	0.73	0.69	0.61	0.66
z	○ 4.8	○ 4.6	○ 4.6	○ 4.2	○ 4.1	○ 3.9	○ 3.9



Lines: electrode pairs with a marked magnitude squared coherence, coloured by z. Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

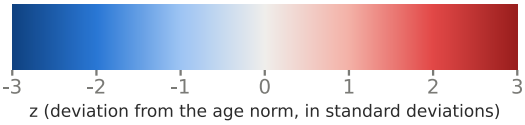
XI. Source power per brain region (sLORETA)

Figure 18: z per brain region and band

Region	Eyes open						Eyes closed					
	delta	theta	alpha	SMR	beta1	beta2	delta	theta	alpha	SMR	beta1	beta2
left frontal	0.3	0.6	0.6	0.6	0.4	0.6	0.1	0.3	0.3	0.3	0.4	0.4
right frontal	0.3	0.5	0.6	0.6	0.4	0.6	0.2	0.3	0.3	0.3	0.4	0.4
left central	0.2	0.1	0.1	0.2	0.0	0.2	0.0	-0.2	-0.1	-0.2	0.0	0.1
right central	0.3	0.1	0.2	0.4	0.1	0.3	0.1	-0.2	-0.1	0.0	0.1	0.1
left temporal	0.4	0.2	0.1	0.2	0.1	0.2	0.1	-0.3	-0.1	-0.1	0.1	0.1
right temporal	0.5	0.2	0.2	0.4	0.2	0.3	0.1	-0.3	-0.1	0.1	0.1	0.1
left parietal	0.0	-0.8	-0.1	-0.1	-0.3	-0.2	-0.2	-1.1	-0.2	-0.3	-0.2	-0.3
right parietal	0.1	-0.7	0.0	0.1	-0.2	-0.1	-0.2	-1.0	-0.1	-0.2	-0.2	-0.2
left occipital	0.0	-1.1	-0.1	-0.1	-0.4	-0.3	-0.3	-1.3	-0.1	-0.3	-0.2	-0.3
right occipital	0.1	-1.1	0.0	0.0	-0.3	-0.2	-0.3	-1.3	-0.1	-0.3	-0.2	-0.3

No value of the estimated source power per brain region, eyes open, is marked.

No value of the estimated source power per brain region, eyes closed, is marked.



Synthetic values: log10 of the estimated source power per region and band, compared with the age norm. Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Source power per brain region is estimated with sLORETA from 19 scalp electrodes on a template head; it does not show where in the brain a signal arises.

XII. Event-related potentials at the electrodes (channel ERPs)

Figure 19: Channel ERPs, Go trials

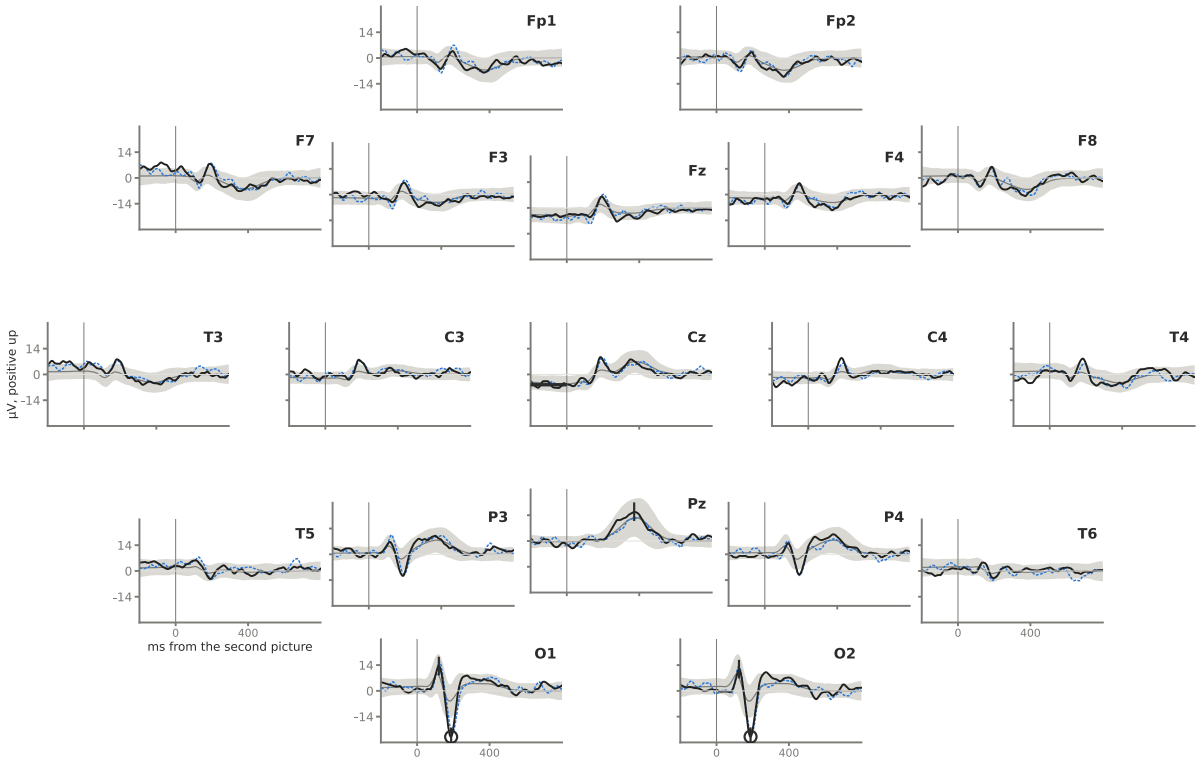


Figure 20: Channel ERPs, NoGo trials

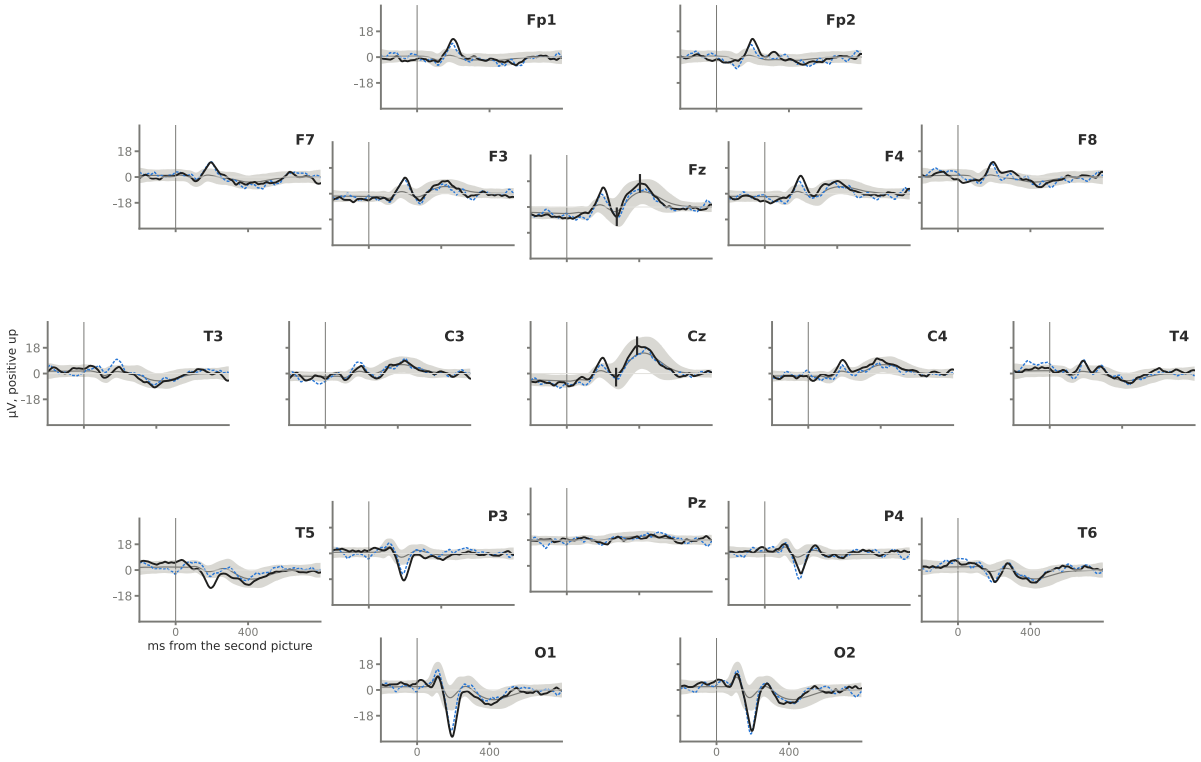


Figure 21: Channel ERPs, Novelty trials

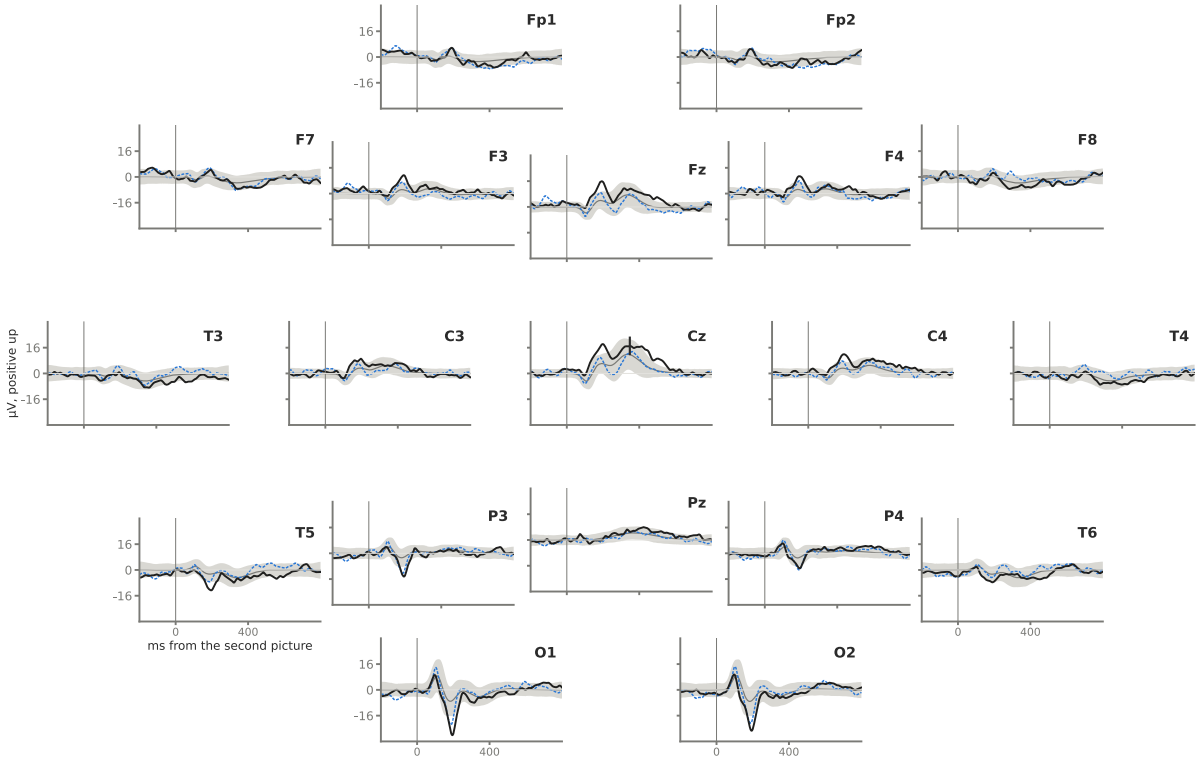


Figure 22: Go and NoGo trials and the difference Go minus NoGo (no norm)

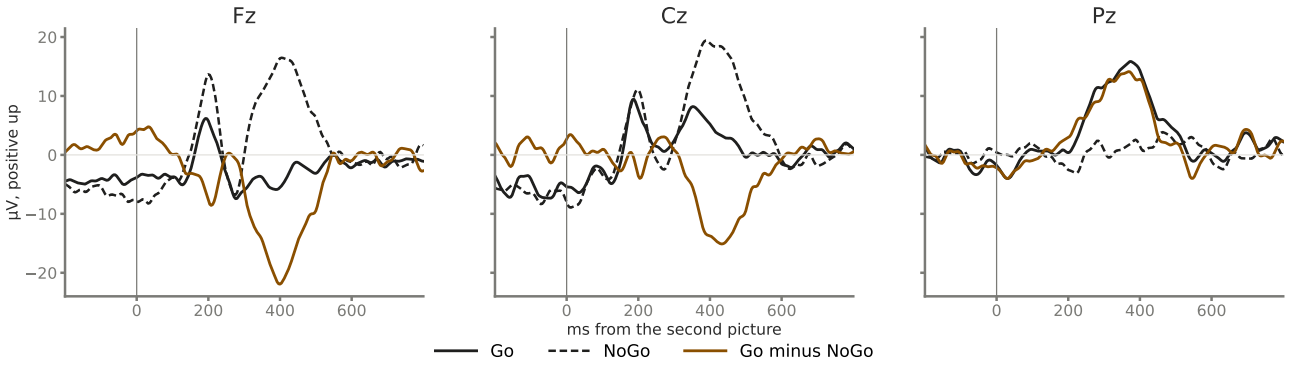


Figure 23: ERP peak values

Component	Trial type	Electrode	Amplitude (µV)	z	Latency (ms)	z
P1	Go	O1	13.5	0.2	120	0.5
P1	Go	O2	11.8	-0.2	124	0.7
N1	Go	O1	-25.0	○ **-4.2**	188	0.3
N1	Go	O2	-24.9	○ **-4.2**	188	0.3
N2	NoGo	Fz	-6.8	0.0	276	0.3
N2	NoGo	Cz	-2.5	0.3	272	0.3
P3	Go	Pz	15.9	0.1	372	-0.1
P3	NoGo	Cz	19.4	0.3	388	-0.7
P3	NoGo	Fz	16.5	0.8	404	-0.3
P3a	Novelty	Cz	17.1	0.5	348	0.4
CNV	Go	Cz	-5.6	-0.5		

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

For N1, N2 and CNV a negative z means a larger negativity. CNV: mean amplitude from -200 to 0 ms before the second picture; no latency.

Black: this measurement. Grey line and band: normative mean and mean ± 2 SD. Time 0: second picture. Tick: scored peak; ring: marked value; bar: CNV window at its mean. Blue dashed: earlier measurement.

ERP peaks, Go trials: the suggested peaks were confirmed unchanged by the clinician.

ERP peaks, NoGo trials: the suggested peaks were confirmed unchanged by the clinician.

ERP peaks, Novelty trials: the suggested peaks were confirmed unchanged by the clinician.

N1 amplitude at O1, Go trials, is more negative than the age norm (-25.0 µV, z = -4.2).

N1 amplitude at O2, Go trials, is more negative than the age norm (-24.9 µV, z = -4.2).

XIII. ERP peak review

Component	Trial type	Electrode	Peak clear	Amplitude	z	Latency	z	Unusual curve shape
P1	Go	O1	yes (11.1)		0.2		0.5	
P1	Go	O2	yes (11.0)		-0.2		0.7	
N1	Go	O1	yes (27.8)	more negative	○ **-4.2**		0.3	
N1	Go	O2	yes (27.8)	more negative	○ **-4.2**		0.3	
N2	NoGo	Fz	yes (13.1)		0.0		0.3	
N2	NoGo	Cz	yes (9.6)		0.3		0.3	
P3	Go	Pz	yes (4.0)		0.1		-0.1	
P3	NoGo	Cz	yes (15.5)		0.3		-0.7	
P3	NoGo	Fz	yes (14.4)		0.8		-0.3	
P3a	Novelty	Cz	no (1.1)		0.5		0.4	
CNV	Go	Cz	window mean (5.0)		-0.5	-	-	

Ticks against the age norm: proposed by the system where the peak is clear and the value is marked, then confirmed or changed by the clinician; the curve shape only the clinician marks. For N1, N2 and CNV a more negative amplitude is a larger negativity. Peak clear: in brackets the peak's prominence over the SD of the residual noise (clear from 2); CNV is a window mean.

ERP peak review, N1 at O1, Go trials: amplitude more negative than the age norm (z = -4.2), proposed by the system and confirmed by the clinician.

ERP peak review, N1 at O2, Go trials: amplitude more negative than the age norm (z = -4.2), proposed by the system and confirmed by the clinician.

ERP peak review, P3a at Cz, Novelty trials: peak not clear; no deviation proposed (amplitude z = 0.5).

ERP peak review: 9 of 11 scored peaks reviewed by the clinician without a deviation tick.

For each scored peak the system proposes a deviation tick only where the peak is clear (a local extremum of the expected polarity inside its window, not at the window edge, with a prominence of at least twice the residual noise) and its amplitude or latency is marked; the clinician confirms or changes the ticks and may mark an unusual curve shape. Where a peak is not clear, nothing is proposed.

XIV. VCPT task performance

Measure	Value	Percentile	z
Omission errors	2	46	-0.1
Commission errors	4	81	0.9
Mean reaction time	409 ms	47	-0.1
Reaction time variability (SD)	86 ms	42	-0.2

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 24: Errors per block of 40 trials (no norm)

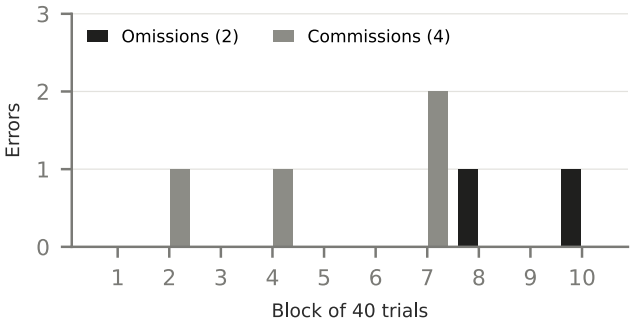


Figure 25: Reaction time over trials

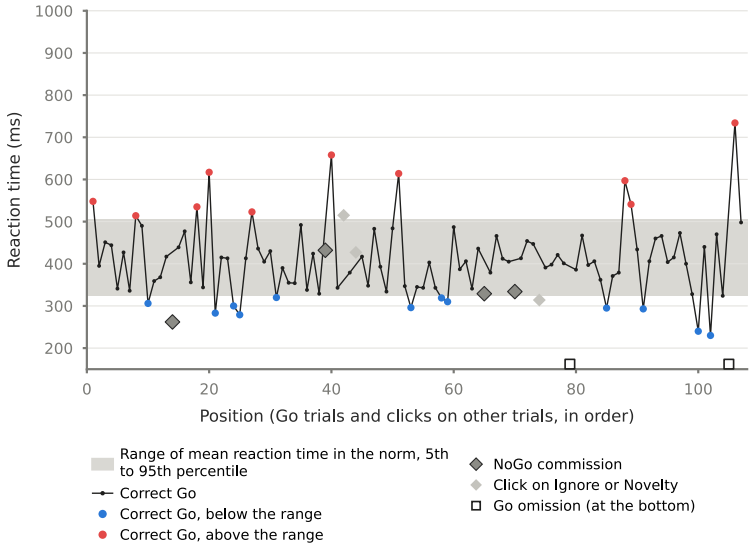
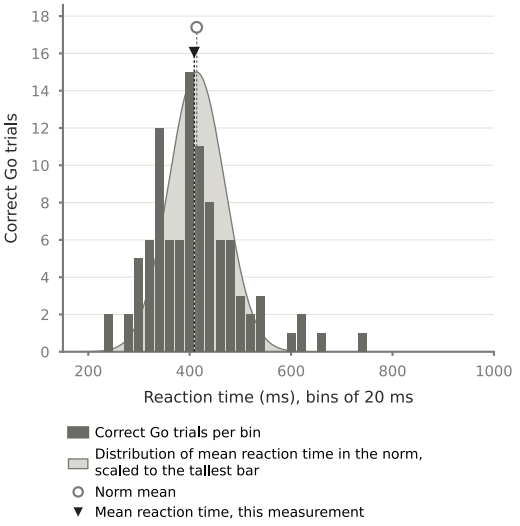


Figure 26: Reaction times of correct Go trials



Reaction time: from the second picture to the click. The band and the curve show how the mean reaction time varies between the people of the norm sample, not how single trials vary.

Omission errors: 2, not marked (percentile 46).

Commission errors: 4, not marked (percentile 81).

Mean reaction time: 409 ms, not marked (percentile 47).

Reaction time variability (SD): 86 ms, not marked (percentile 42).

XV. Comparison of measurements

Measure	Earlier 23 Sep 2025 (uploaded later)	z	Earlier 12 Mar 2026	z	Current 15 Sep 2026	z
Omission errors	3	0.3	3	0.3	2	-0.1
Commission errors	6	1.5	5	1.2	4	0.9
Mean reaction time	431 ms	0.2	421 ms	0.1	409 ms	-0.1
Reaction time variability (SD)	97 ms	0.2	95 ms	0.2	86 ms	-0.2
Theta/beta ratio at Cz, Eyes open	2.54	-0.3	2.22	-0.7	2.54	-0.3
Alpha peak frequency (Hz), Eyes open	9.1	0.1	9.1	0.0	9.1	0.0
Theta/beta ratio at Cz, Eyes closed	2.47	-0.4	2.26	-0.7	2.66	-0.2

Alpha peak frequency (Hz), Eyes closed	9.3	0.3	9.2	0.0	9.2	0.0

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 27: Earlier 23 Sep 2025: Relative power, z against the age norm, Eyes open

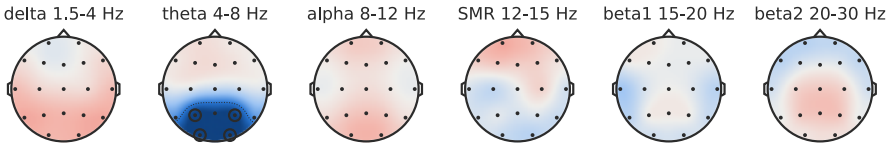


Figure 28: Earlier 12 Mar 2026: Relative power, z against the age norm, Eyes open

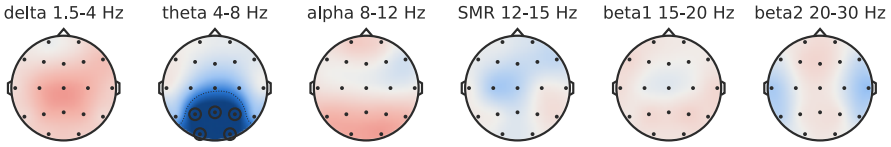
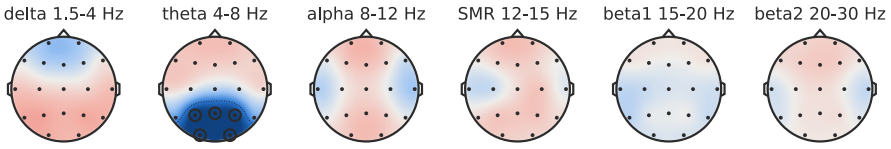


Figure 29: Current 15 Sep 2026: Relative power, z against the age norm, Eyes open



Ring: marked value (after correction for multiple comparisons).

Beta1 coherence, eyes closed, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: C3-P3 4.5, 5.0 and 3.9; T3-T5 4.4, 4.7 and 4.8; C3-T5 4.5, 4.8 and 3.9; T3-C3 4.5, 4.7 and 4.1; T5-P3 4.1, 4.5 and 4.6; T5-O1 4.4, 4.6 and 4.6; P3-O1 4.3, 4.4 and 4.2.

Beta1 coherence, eyes open, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: T3-C3 4.1, 4.8 and 4.1; C3-T5 4.8, 4.3 and 3.8; T3-T5 4.2, 4.5 and 3.8; T5-P3 4.3, 4.5 and 3.7; T5-O1 4.3, 4.3 and 4.1; C3-P3 4.3, 4.3 and 3.7; P3-O1 4.2, 4.2 and 4.2.

N1 amplitude, Go, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: O2 -4.4, -3.9 and -4.2; O1 -4.2, -3.7 and -4.2.

SMR asymmetry index, eyes open, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: C3-C4 4.1, 3.4 and 4.3.

SMR asymmetry index, eyes closed, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: C3-C4 3.7, 3.8 and 4.1.

Beta2 coherence, eyes open, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: P4-O2 0.7, 0.5 and 3.9.

Relative theta power, eyes open, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: P4 -3.3, -3.7 and -3.4; P3 -3.4, -3.6 and -3.5; Pz -3.0, -3.6 and -3.2; O1 -3.4, -3.3 and -3.4; O2 -3.3, -3.3 and -3.3.

This comparison lists 24 of the 24 values marked in at least one of the measurements, ranked by |z|: the 6 largest and every value marked at 3 z or more.

The earlier measurement of 23 Sep 2025, uploaded later, was processed with the current software and norms (V4 mockup 0.1, mockup pipeline 0.1.0; Mock norms 2026.1 (synthetic)) as result set RS-MOCK-042-0923-1.

The earlier measurement of 12 Mar 2026 was processed again with the current software and norms (V4 mockup 0.1, mockup pipeline 0.1.0; Mock norms 2026.1 (synthetic)) as result set RS-MOCK-042-0312-2; result set RS-MOCK-042-0312-1, from which report R-MOCK-042-0312 v1 was issued, stays unchanged.

Measurements are shown side by side, each against its own age norm; differences between them are not tested for statistical significance.

Measurements uploaded later are processed from their recordings; reports and numbers made by earlier software are not imported.

XVI. Notes on the results

The results are one input to the clinician's assessment. The software does not diagnose and does not recommend treatment.

z is the distance of a value from the normative mean, in standard deviations of the normative sample; about 95% of the normative sample lies between z = -2 and z = 2.

Marked values differ from the age norm after correction for multiple comparisons, which keeps chance results among the many comparisons rare; a value beyond 2 z that is not marked may deviate by chance alone. The correction uses a false discovery rate method; how it accounts for correlated values is decided at implementation.

XVII. Method and norms

Recording: Mitsar amplifier, WinEEG file (mockup label); 19 scalp electrodes of the 10-20 system (Fp1 to O2); conditions: eyes open, eyes closed and VCPT.

Processing: average reference of the 19 scalp channels; band-pass filter 0.5 to 45 Hz and mains notch filter at 50 Hz; analysis rate 250 Hz; artefacts removed by independent component analysis (ICA) with reference maps, then remaining artefact segments excluded.

Spectra: 1 to 30 Hz in steps of 0.5 Hz from windows of 2 s with 50 % overlap, compared with the norm as log10 power; relative power is the share of the total power from 1.5 to 30 Hz. Bands: delta 1.5 to 4 Hz, theta 4 to 8 Hz, alpha 8 to 12 Hz, SMR 12 to 15 Hz, beta1 15 to 20 Hz and beta2 20 to 30 Hz. Theta/beta ratio at Cz with theta 4 to 8 Hz and beta 13 to 21 Hz.

ERPs: VCPT with 100 trials each of Go, NoGo, Ignore and Novelty; averages of correct trials only; epochs from -200 to 800 ms around the second picture, baseline before the cue; peaks read off the curves in fixed windows (P1 70 to 170 ms, N1 140 to 250 ms, N2 210 to 330 ms, P3 (Go) 280 to 560 ms, P3 (NoGo) 300 to 600 ms, P3a 250 to 420 ms); CNV as the mean from -200 to 0 ms.

Norms: Mock norms, version 2026.1 (synthetic): one continuous model of age and sex per measure (stand-in for the continuous GAMLSS model proposed for the core), fitted to 400 synthetic subjects per condition; age range 6 to 58 years.

Marking: a value is marked when it differs from the age norm after correction for multiple comparisons with a false discovery rate method; this mockup applies Benjamini-Hochberg (q = 0.05) within each analysis block as a stand-in until the handling of correlated values is decided.

Recording conditions (documented at upload)

Time of recording	09:40
Medication on the recording day	none reported
Time of the last dose	not applicable
Sleep the night before	as usual
Corrected vision (glasses or lenses)	not documented
Response hand (VCPT)	not documented
Technician's note: drowsiness during the recording	no

Issued by

Issued by	_____ (entered when the clinician issues the report)
Date of issue	_____ (entered when the clinician issues the report)
Signature	_____ (entered when the clinician issues the report)
Version	1 · Draft, not issued
Replaces	none, first version

Text sources

System sentences in this report come from the fixed sentence library 2026.2. Sentence ids by section:

Summary `SUM-2, SUM-RULE-3, SUM-COUNT-1`

Signal quality and processing `Q-DUR-1, Q-TRIALS-1, Q-ICA-MAN-1, Q-SEG-AUTO-1, MK-AUTO-1`

Spectra and maps `SP-REL-1, APF-IN-2, SP-NONE-2`

Frequency markers `FM-AUTO-1, FM-RULE-1`

Band ratios and indices `TBR-1, FAA-1, FTPA-1, AR-1`

Asymmetry `ASY-1`

Coherence `COH-MSC-1`

Source power per brain region (sLORETA) `SRC-NONE-1, SRC-LIM-1`

Event-related potentials at the electrodes (channel ERPs) `ERP-PEAK-AUTO-1, ERP-AMP-NEG-1`

ERP peak review `ERPREV-SUG-1, ERPREV-UNCLEAR-1, ERPREV-REST-1, ERPREV-RULE-1`

VCPT task performance `BEH-IN-2`

Comparison of measurements `CMP-G-1, CMP-RULE-1, L-REPROC-2, L-REPROC-1, L-SIDE-1, L-IMPORT-1`

Notes on the results `L-AID-1, L-Z-1, L-MC-1`

Method and norms `MET-REC-1, MET-PROC-1, MET-SPEC-1, MET-ERP-1, MET-NORM-1, MET-MC-1`